Mosairul Alam Tyhan

Aspiring Machine Learning Engineer

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PROFESSIONAL SUMMARY

Aspiring **Machine Learning Engineer** with hands-on experience developing and deploying end-to-end ML applications using **Python, Scikit-learn, TensorFlow, FastAPI, Docker, and Streamlit**. Built **NLP applications**, including **NewsCortex AI**, a news classification system trained on **120K+ articles with 91% accuracy**, with experience in model development, REST APIs, and deployment. Currently seeking an **ML internship or entry-level opportunity** to apply my skills and gain practical experience in real-world machine learning projects.

EXPERIENCE

Machine Learning Intern Jul 2026 - Present
FlyRank *Remote*

- Participating in **ML-focused training and practical sessions** covering AI fluency, AI tools, and problem-solving approaches.
- Completing **hands-on individual assignments** involving problem understanding, information gathering, analysis, and solution development.
- Applying learned concepts by **building and experimenting with AI/ML projects**, using AI tools to support research, learning, and technical problem-solving.

TECHNICAL SKILLS

Languages :	Python, SQL
ML/DL :	Scikit-learn, TensorFlow, Keras, Numpy, Pandas
Backend & Deployment :	FastAPI, Docker, Streamlit, Uvicorn, REST APIs, Streamlit Community Cloud
Tools & Database :	Git, GitHub, Jupyter, Kaggle, MySQL
ML Areas :	NLP, Deep Learning, Model Deployment

PROJECTS

Freight Rate Predictor | 2026

- Built an **end-to-end freight rate prediction system**, including preprocessing, feature engineering, and regression model comparison.
- Developed a **FastAPI backend and Streamlit interface** with input validation for real-time freight-rate predictions.
- Containerized and deployed the application with **Docker**, with a documented **video walkthrough** explaining the complete approach and implementation.

NewsCortex AI | 2026

- Built an **NLP-based news classification system** using TF-IDF, TensorFlow/Keras, and ANN (Deep Learning).
- Processed 120K+ articles across 4 categories with 10K TF-IDF features using **unigram/bigram representations**.
- Achieved 91% test accuracy and built a **FastAPI backend, Dockerized** the application, and **deployed a Streamlit frontend** for real-time predictions.

Heart Disease Risk Predictor | 2026

- Built a binary classification model using **Logistic Regression** to predict heart disease risk.
- Developed a **FastAPI REST API** with **Pydantic validation** for real-time predictions and probability scores.
- **Dockerized and deployed** the application with a **Streamlit frontend**, featuring feature-importance visualization and error handling.

EDUCATION

Diploma in Engineering in Computer Science & Technology
Mymensingh Polytechnic Institute
Expected Graduation: 2027

CERTIFICATION

Data Science & Machine Learning with Python
Ostad
Data Science Workshop for Research & Higher Studies
Ostad

LANGUAGES

- Bangla (Native)
- English (Working Proficiency)